

Logistic Regression (Binary Classification) ←

Dataset

Pass / Fail
o/p & Binary categories

Study hours	Pass / Fail
2	Fail
3	Fail
4	Fail
5	Pass
6	Pass
7	Pass
8	Pass

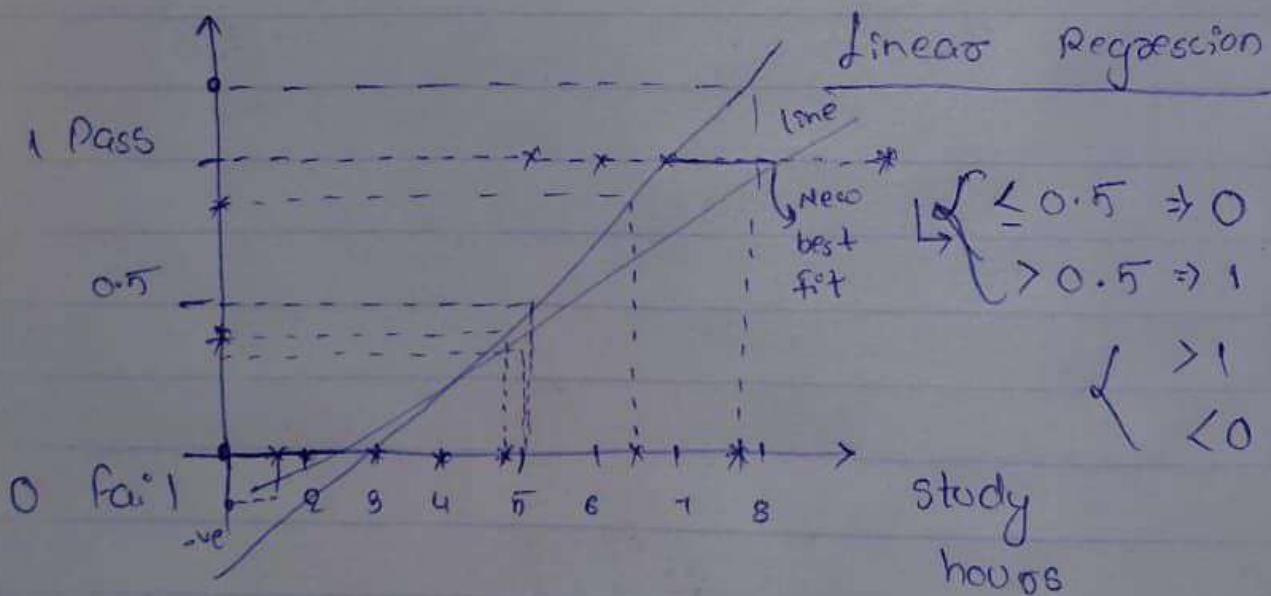
Study of hours



o/p

Pass / Fail

Logistic Regression
0 to 1



$$\begin{cases} \leq 0.5 \Rightarrow 0 \\ > 0.5 \Rightarrow 1 \end{cases}$$

$$\begin{cases} > 1 \text{ and } < 0 \end{cases}$$

Why we cannot use Linear Regression for classification?

- 1) Outlier & Best fit line change
- 2) > 1 and < 0 (squash line)